

ROSTOV, Russian Federation

WMO: 273290

Lat: 57.200N

Lon: 39.417E

Elev: 330

StdP: 14.52

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.7	-9.8	-19.9	1.9	-15.5	-13.9	2.6	-9.5	16.9	26.9	14.7	25.9	2.5	350	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	84.6	68.6	81.2	67.6	78.0	66.1	71.2	81.3	69.2	78.6	67.3	75.9	4.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	102.5	76.8	65.6	95.6	74.5	63.9	89.9	72.5	35.2	81.4	33.5	79.1	32.0	76.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
14.5	12.6	11.2	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
			WB	-21.5	87.7	5.4	3.5	-25.4	90.2	-28.6	92.3	-31.6	94.2	-35.6	96.8	
				-21.7	73.4	5.4	2.4	-25.6	75.1	-28.8	76.6	-31.8	78.0	-35.7	79.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	16.1	17.0	26.7	41.6	53.9	61.1	66.3	61.4	51.9	40.7	29.0	21.1	
	DBStd	19.88	13.08	13.17	9.21	8.89	8.73	6.89	6.23	6.52	6.90	7.94	9.80	12.48	
	HDD50	4924	1050	924	723	280	59	3	0	2	57	301	630	895	
	HDD65	9059	1515	1344	1188	702	356	157	60	150	394	753	1080	1360	
	CDD50	1534	0	0	0	29	181	336	504	354	116	14	0	0	
	CDD65	192	0	0	0	0	13	40	99	37	3	0	0	0	
	CDH74	1729	0	0	0	3	126	368	868	349	15	0	0	0	
	CDH80	428	0	0	0	0	27	80	224	97	0	0	0	0	
Wind	WSAvg	5.0	5.5	5.5	5.5	5.1	5.2	4.5	4.0	4.1	4.2	5.2	5.6	5.7	
Precipitation	PrecAvg	23.1	1.6	1.3	1.1	1.3	2.0	2.7	2.5	2.8	2.3	2.2	1.8	1.7	
	PrecMax	32.1	2.4	2.9	2.7	2.5	3.9	6.3	6.4	7.2	5.1	5.5	3.3	3.0	
	PrecMin	15.2	0.6	0.2	0.2	0.5	0.6	0.0	0.2	0.6	0.8	0.6	0.1	0.4	
	PrecStd	3.9	0.5	0.6	0.6	0.5	0.7	1.3	1.4	1.6	1.1	1.3	0.9	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.8	38.9	53.1	72.4	83.7	86.3	90.0	88.7	76.1	64.9	50.6	44.7	
		MCWB	36.3	35.6	43.2	56.0	65.6	68.3	72.0	71.4	65.2	56.1	47.4	43.0	
	2%	DB	35.3	36.4	45.3	66.8	77.7	82.2	85.2	81.8	70.9	58.4	45.8	40.4	
		MCWB	33.8	34.0	38.2	53.1	63.3	67.0	69.4	68.5	61.4	52.6	43.9	39.1	
	5%	DB	34.1	34.9	40.8	61.4	72.8	78.5	82.0	77.4	67.0	54.5	43.2	36.9	
		MCWB	33.0	33.2	35.9	50.2	59.4	65.8	68.7	66.2	59.0	50.4	41.5	35.5	
	10%	DB	32.8	33.6	38.0	56.3	69.1	74.6	78.8	73.4	63.3	51.5	40.4	34.7	
		MCWB	31.7	32.3	34.6	47.2	57.3	63.7	67.4	63.7	57.2	48.8	39.1	33.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	36.6	44.3	57.2	69.1	71.3	74.4	73.5	66.3	57.3	48.1	43.4
			MCDB	37.5	38.1	52.6	70.1	80.6	81.9	84.8	86.6	73.5	63.3	50.1	44.5
2%		WB	34.2	34.7	39.0	54.1	64.8	68.9	72.1	70.1	62.9	53.8	44.3	39.4	
		MCDB	34.9	35.8	43.0	64.6	75.2	79.0	82.3	79.6	69.0	57.9	45.6	40.3	
5%		WB	33.2	33.5	36.7	51.3	61.7	67.1	70.0	67.5	60.1	51.2	41.6	35.8	
		MCDB	34.0	34.6	39.9	60.0	70.5	76.3	79.6	75.4	65.4	53.9	42.9	36.6	
10%		WB	31.7	32.3	35.1	48.3	59.0	64.8	68.2	65.0	57.7	48.8	39.1	33.6	
		MCDB	32.8	33.7	37.5	55.4	66.0	72.4	76.8	71.3	62.4	51.1	40.4	34.4	
Mean Daily Temperature Range	5% DB	MDBR	10.4	12.8	14.5	16.7	18.1	17.6	17.9	17.4	15.3	10.9	7.5	8.5	
		MCDBR	6.6	9.3	17.2	23.9	23.6	22.9	22.4	23.8	21.3	16.5	9.8	6.6	
		MCWBR	6.6	8.2	11.6	12.9	11.4	10.5	9.9	11.6	12.1	10.7	8.7	6.3	
	5% WB	MCDBR	6.6	8.2	14.0	21.9	20.8	20.4	20.0	21.5	18.8	13.9	8.9	6.3	
		MCWBR	6.7	7.7	9.6	12.3	11.2	10.3	10.1	11.5	11.2	10.0	8.3	6.2	
Clear-Sky Solar Irradiance	taub	0.284	0.326	0.346	0.403	0.384	0.374	0.412	0.417	0.372	0.350	0.314	0.278		
	taud	2.088	2.063	2.088	2.143	2.334	2.381	2.300	2.269	2.369	2.337	2.210	2.078		
	Ebn at Noon	177	218	250	254	270	275	260	249	244	214	166	142		
	Edh at Noon	18	29	37	41	36	35	37	36	28	23	16	14		
All-Sky Solar Radiation	RadAvg	161	431	921	1281	1682	1733	1707	1337	815	364	134	83		
	RadStd	27	72	90	128	141	161	195	130	117	62	23	22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page